

Can Street Networks Predict Neighborhood Change?

Daniel Krasner | URB 8040



Neighborhood Bones

Patterns of Neighborhood Change

Studies on urban dynamics often emphasize the importance of examining neighborhood change over time to uncover spatial and temporal trends. Factors such as economic shifts, demographic transitions, and policy interventions are frequently highlighted as drivers of neighborhood evolution.

Classifying Neighborhood Change

Urban planning literature underscores the value of typologies in making sense of complex urban processes. Data-informed classifications allow researchers to identify distinct neighborhood trajectories, such as gentrification, stability, or decline, which can inform policy and investment decisions.

Static Infrastructure and Predictive Power

The relationship between infrastructural variables and neighborhood change is a growing area of research. Proximity to transit, street network connectivity, and road infrastructure are increasingly studied as stable yet influential factors shaping long-term urban development patterns and economic outcomes. Mixed with the narratives around affordability, livable cities, and movements such as YIMBYism, this is pertinent.

Research Questions, Variables, High-Level Findings

Because we should always start with the findings!



Objective / Research Questions

- To understand **patterns** of neighborhood change in the city of Atlanta from 2018 to 2022.
- To **classify** types of neighborhood change in Atlanta during this time period, defining a data-informed typology.
- To **visualize** neighborhood change.
- To **identify** if more static, infrastructural variables – such as a tracts' proximity to MARTA, network density, or proportion of roads that are highways – have any predictive power in the type of neighborhood change it experienced.

Expected Measures of Change

Stage Setting Variables

- **Vacancy Rates:** High vacancy rates often signal disinvestment, instability, or transitional stages within neighborhoods, setting the foundation for potential future change.
- **Median Home Values:** Lower median home values can indicate affordability challenges, potential for gentrification, or existing economic disadvantage.

Demographic and Socioeconomic Variables (Change Over Time)

- **Non-White Population Percentage:** Helps identify demographic shifts and historical racial disparities in gentrification processes.
- **Median Household Income:** Tracks economic improvement or decline within neighborhoods, providing a lens for understanding wealth redistribution.
- **Education Levels (Bachelor's or Higher):** Reflects shifts in socioeconomic status and workforce composition, often a marker of gentrification or stabilization.
- **Home Value and Income Percentage Changes:** Percentage changes help identify neighborhoods experiencing rapid economic growth or decline, distinguishing stable areas from those undergoing transformation.

Developed Variables of Change

Infrastructure and Accessibility Variables

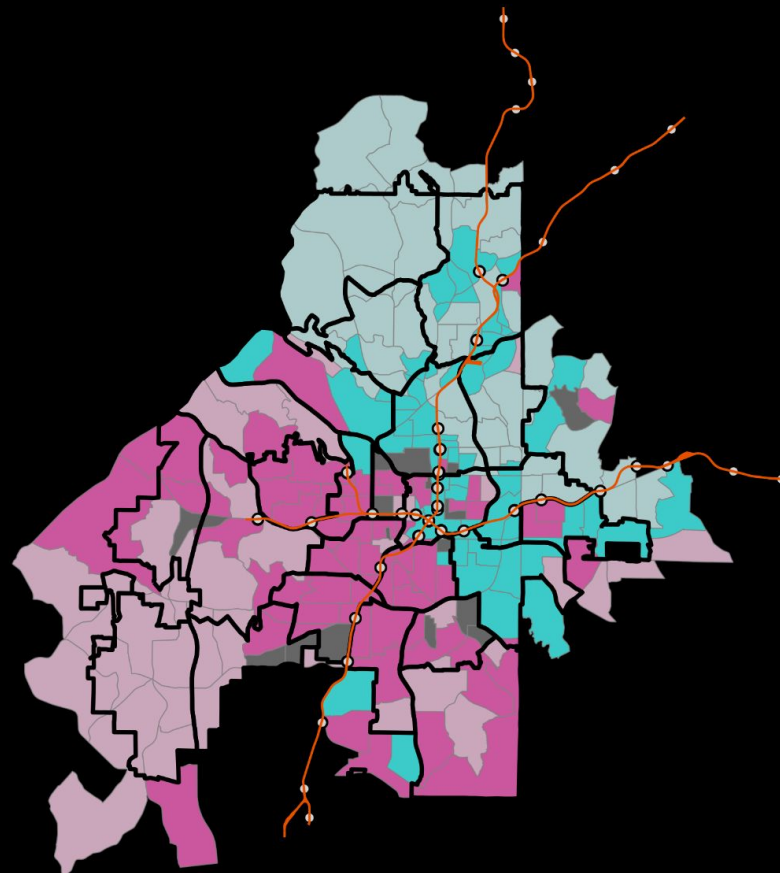
- **Distance to MARTA Stops:** Proximity to public transit hubs reflects accessibility and can drive both investment and desirability in neighborhoods (as the crow flies, not using network).
- **D3B (Intersection Density):** Higher intersection density is a proxy for walkability and "good bones" infrastructure, often associated with urban livability and pedestrian-oriented development.
- **Highways + Ramps Percentage:** A high proportion of roadways dedicated to highways and ramps can suggest auto-oriented design, reducing walkability and potentially limiting neighborhood connectivity.

Clustering Analysis (Neighborhood Typologies)

- **"Historically Disadvantaged, Rapid Gentrification":** Areas with low initial values for home prices and income, showing significant upward change.
- **"Wealthy, Stable Areas":** High home values and incomes, with minimal change over time.
- **"Middle-Class, Moderate Gentrification":** Moderate increases in wealth and socioeconomic status, often transitional.
- **"Low-Income, Slow Gentrification":** Marginal change, remaining affordable but showing early signs of potential transformation.
- **"Unclassified":** Data-missing areas that don't fit into predefined clusters, representing unique contexts.

Findings: Classification Model

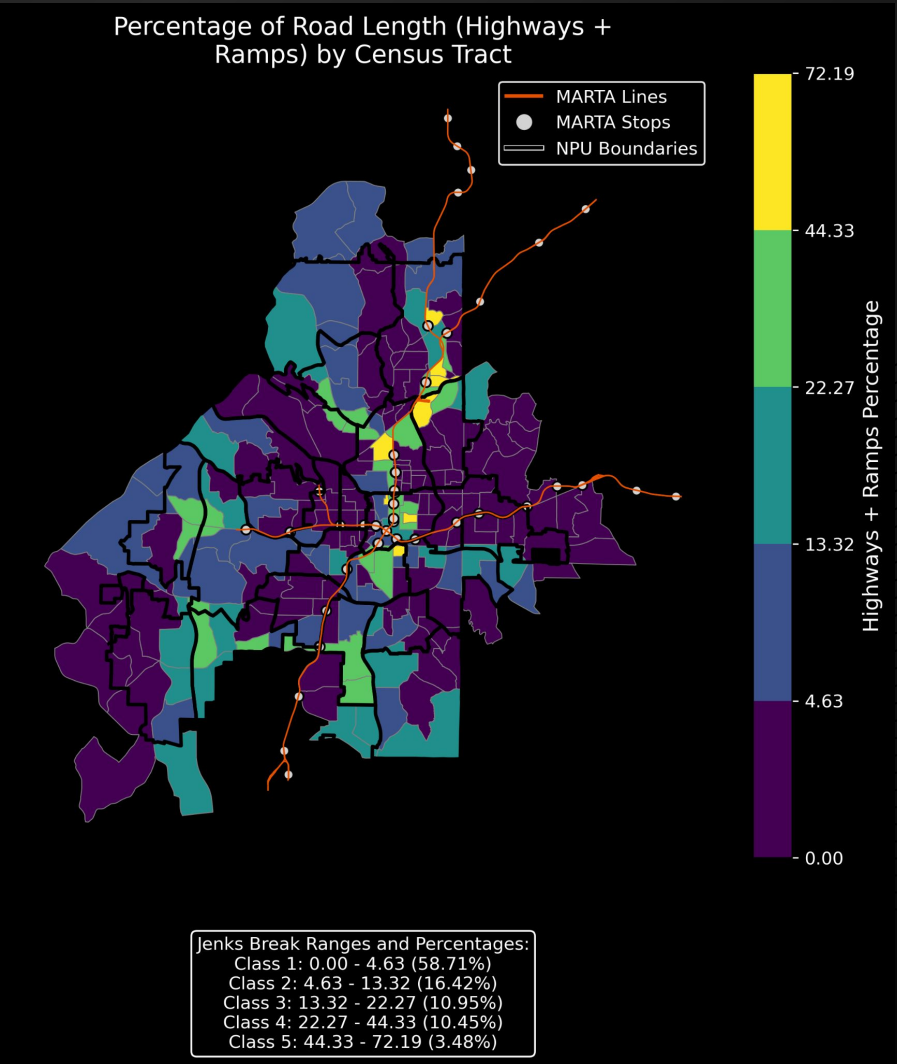
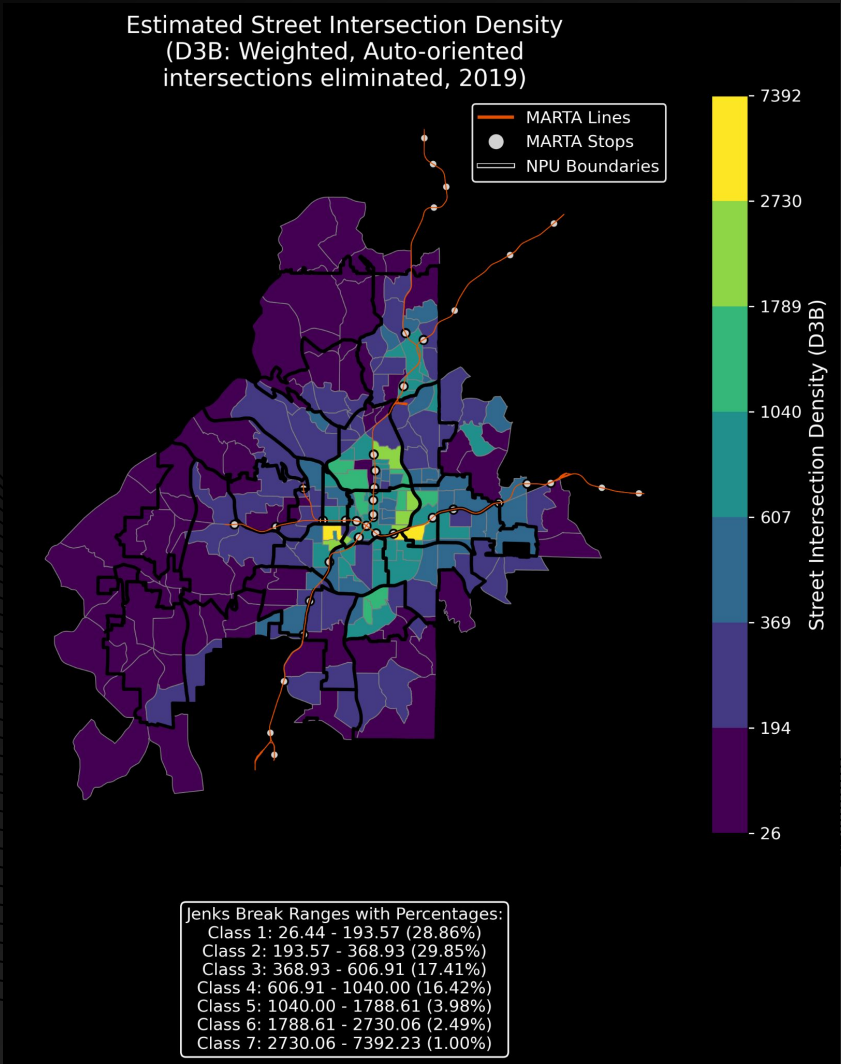
Clustering of Atlanta Census Tracts by
Neighborhood Change



- Teal circle: Middle-Class, Moderate Gentrification
- Pink circle: Historically Disadvantaged, Rapid Gentrification
- Light pink circle: Low-Income, Slow Gentrification
- Light blue circle: Wealthy, Stable Areas
- Grey circle: Data Missing
- Orange line: MARTA Lines
- White circle: MARTA Stops
- Black line: NPU Boundaries

Middle-Class, Moderate Gentrification: 28.36% of tracts
Historically Disadvantaged, Rapid Gentrification: 26.87% of tracts
Wealthy, Stable Areas: 20.40% of tracts
Low-Income, Slow Gentrification: 16.92% of tracts
Unclassified: 7.46% of tracts

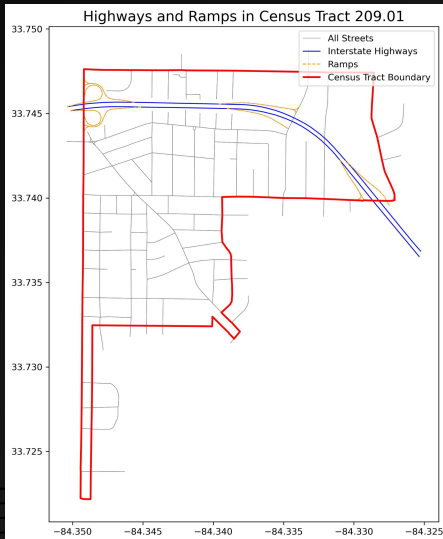
Findings: "Bones" Variables



What was the process to obtaining these findings?

Methodology

Methodologies:



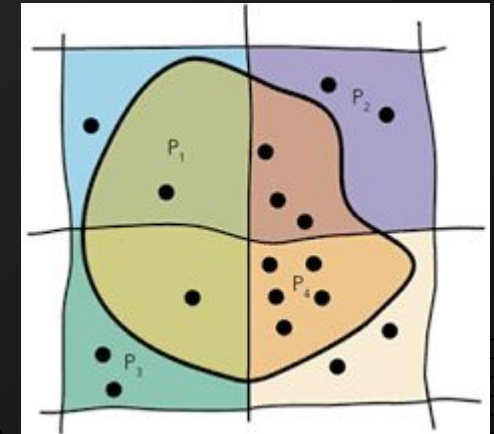
OSMnx

Pulled MARTA stations and network as well as street networks by census tract, identifying highways and ramps.



Geometry Crosswalk

NHGIS crosswalk files allow for apportionment between different geometries in census data.



Proportion Estimates

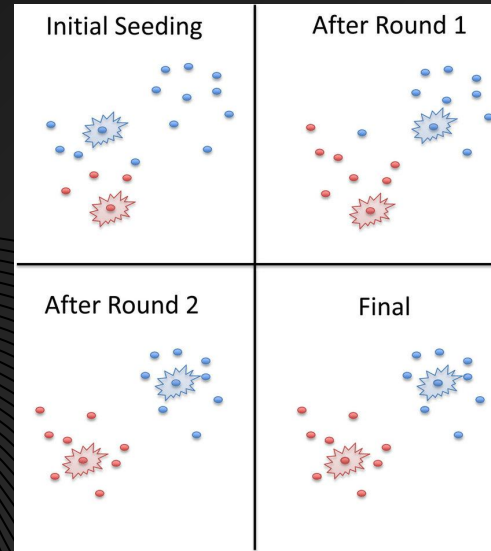
Polygon weighting by proportion specifically for un-crosswalk-able data (walkability index, 2019 block groups, no known cw weight).

Methodologies Continued:



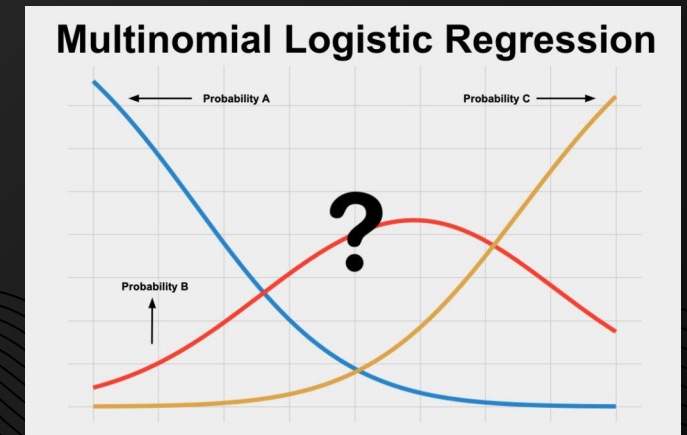
Correlation / OLS

Used correlation tables, OLS / linear regressions to understand basic characteristics and associations.



K Means

Algorithm for data-driven classification of neighborhood change typologies.



Multinomial Regression

Tests predictive power of "static" variables in predicting categories generated by K Means Algorithm.

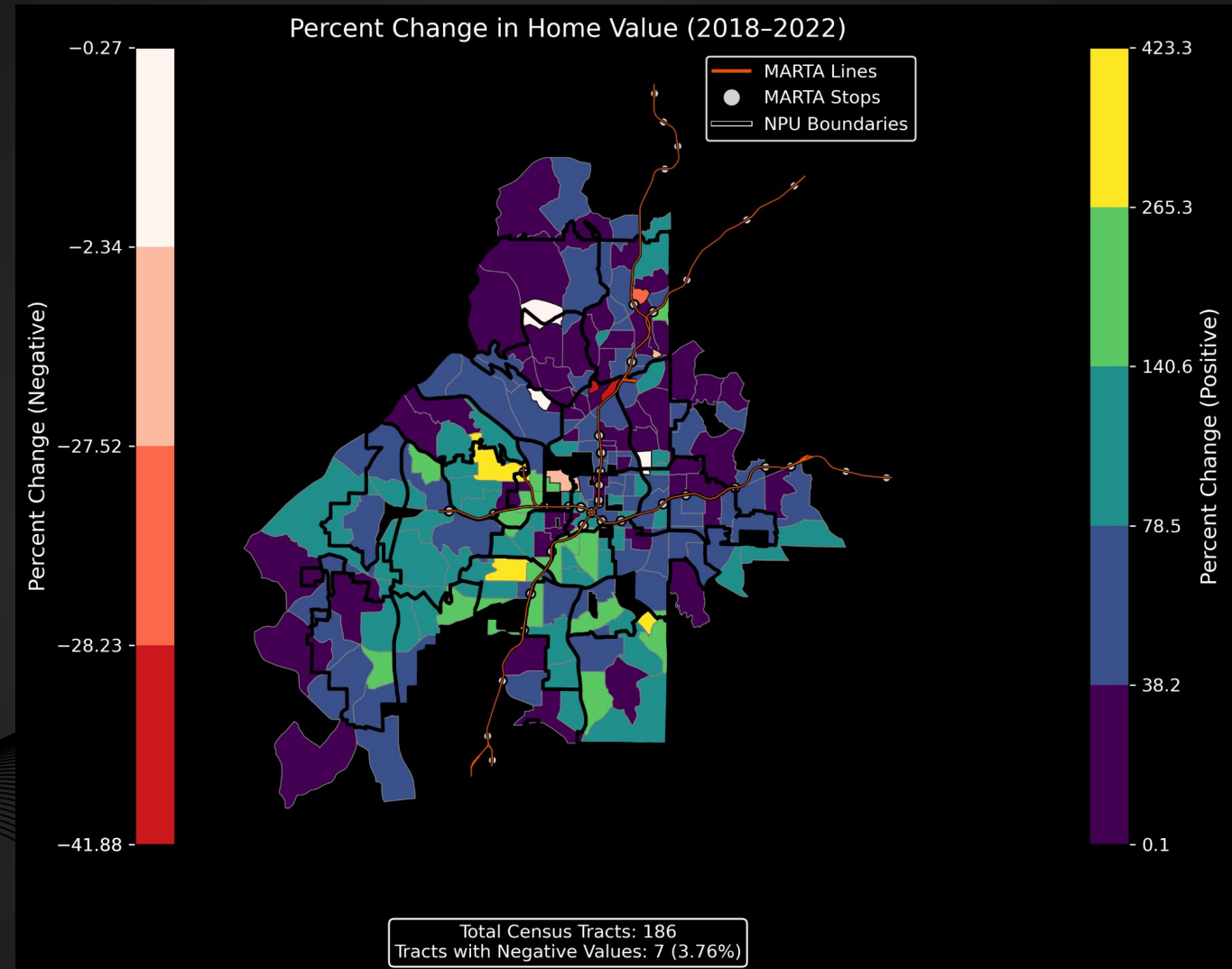
How much did each dynamic variable change from 2018 – 2022?

Including a basic summary statistic description of each distribution of deltas

Percent Change in Home Value

- **Mean (68.20%):** On average, home values increased significantly between 2018 and 2022.
- **Median (53.21%):** The median value is lower than the mean, indicating that there are a few tracts with very high increases (e.g., up to 423.31%) skewing the average upward.
- **Standard Deviation (64.69%):** There is considerable variation in the percent change of home values across tracts, suggesting disparities in property value appreciation.
- **Min (-41.88%):** Some tracts experienced a decline in home values, possibly due to specific economic or social factors.
- **25th percentile (26.67%):** One-fourth of the tracts saw increases below this level.
- **75th percentile (84.78%):** One-fourth of the tracts saw increases above this level.

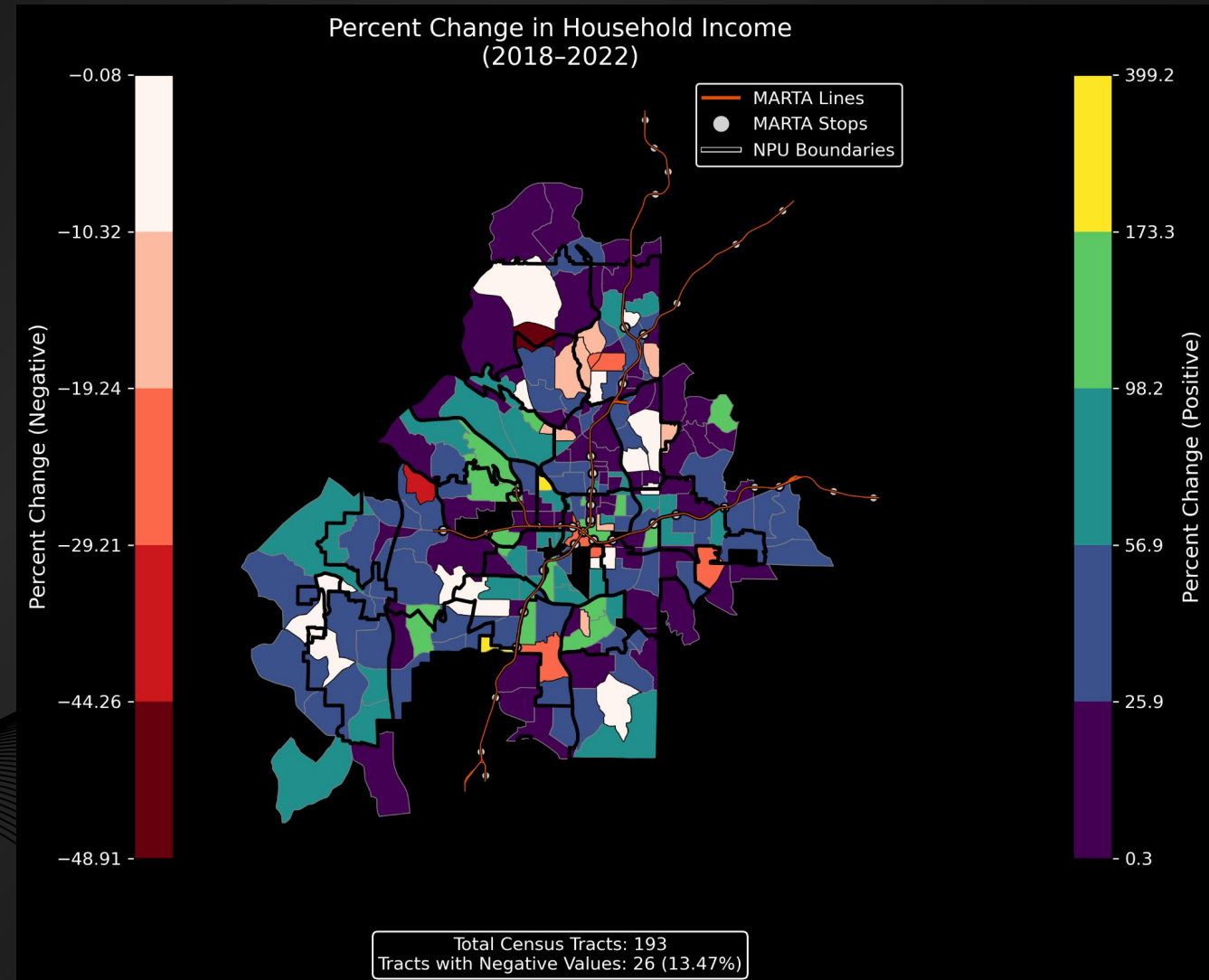
Key Takeaway: The real estate market in Atlanta showed significant growth in many tracts, but the variation highlights underlying disparities. Some areas experienced dramatic increases, likely linked to gentrification or other local economic factors.



Percent Change in Household Income

- **Mean (37.21%):** Average household incomes also grew substantially, but less dramatically than home values.
- **Median (29.21%):** The distribution is right-skewed, with some tracts experiencing very high income growth (max 399.15%).
- **Standard Deviation (46.21%):** There is a wide range of income growth across tracts, indicating uneven economic gains.
- **Min (-48.91%):** Some areas saw declines in median household income, signaling potential economic distress or demographic shifts.
- **25th percentile (8.62%):** One-fourth of the tracts saw modest or negligible growth in income.
- **75th percentile (53.71%):** Income growth was strong in the top quarter of tracts.

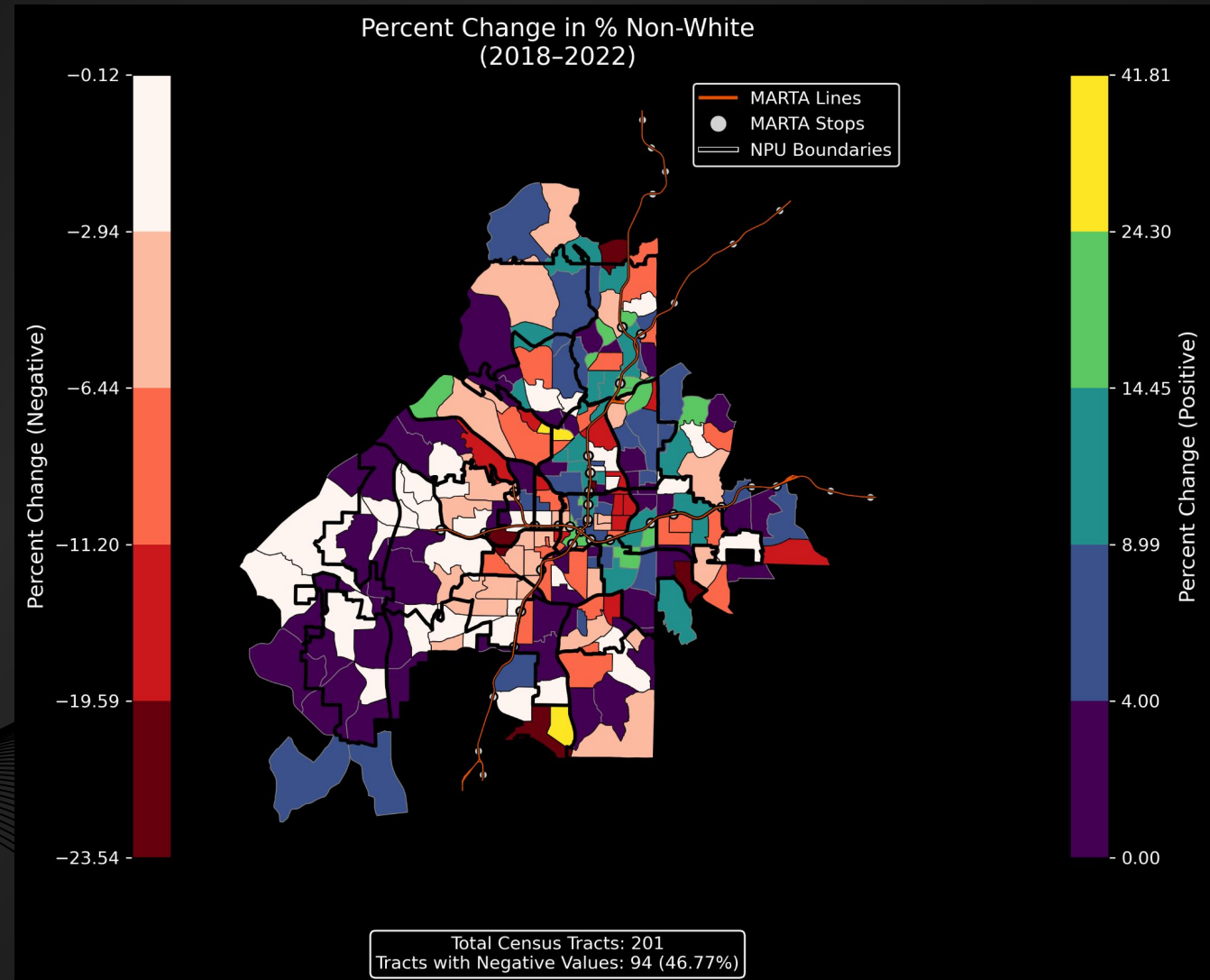
Key Takeaway: Income growth occurred unevenly, with some areas thriving and others stagnating or declining. This may reflect economic polarization and changes in the composition of households.



Percent Change in % Non-White Population

- **Mean (0.77%):** On average, the percentage of non-white residents remained stable, but the standard deviation (9.35%) suggests significant variation.
- **Median (0.43%):** The slight positive median suggests that many areas had minimal changes in racial composition.
- **Standard Deviation (9.35%):** Large shifts occurred in some tracts, with some seeing increases of up to 41.81%, while others saw declines of -23.54%.
- **25th percentile (-4.42%):** Some areas saw reductions in the non-white population.
- **75th percentile (5.88%):** Others experienced modest growth in non-white residents.

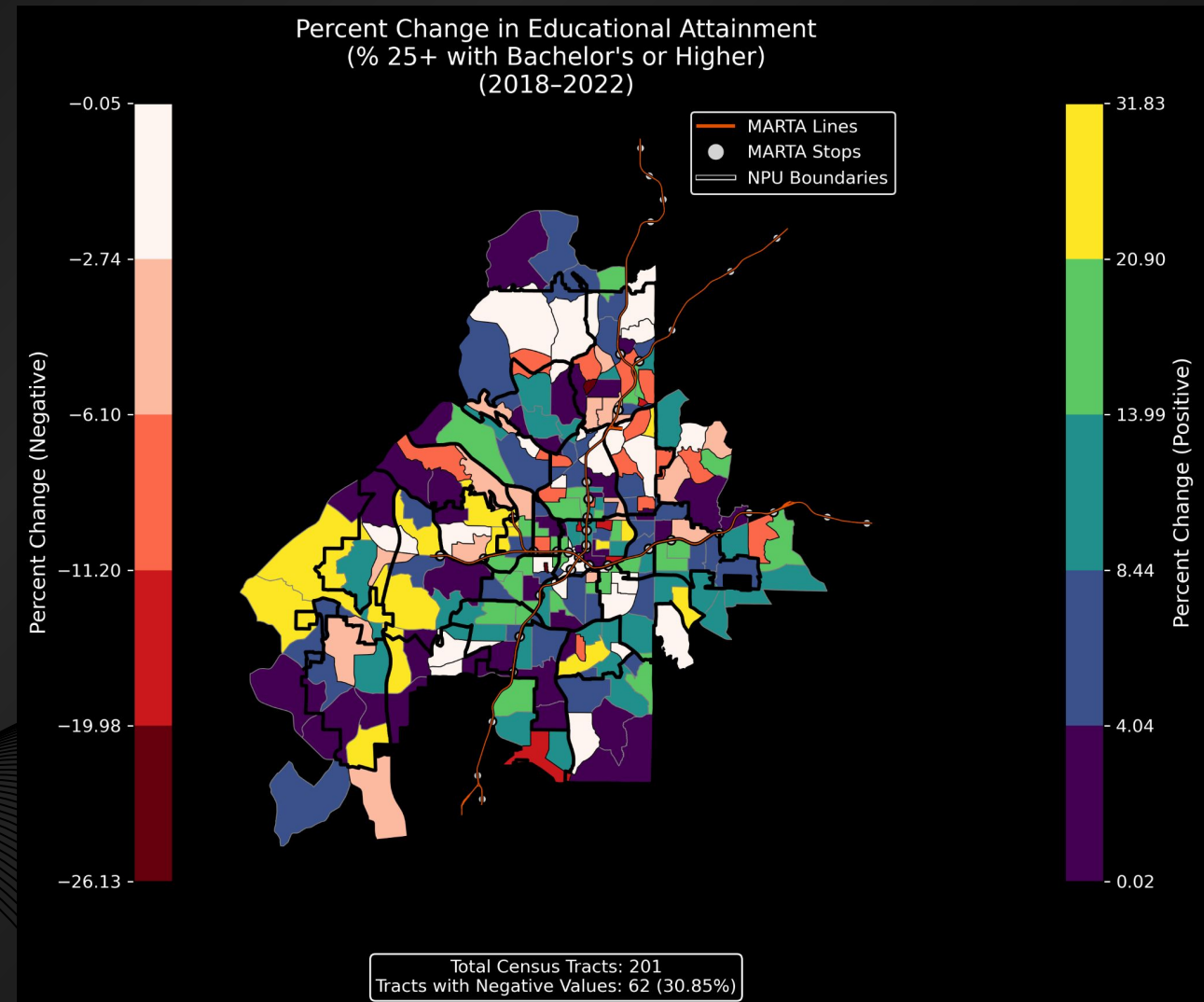
Key Takeaway: While the overall racial composition of the city may have stayed relatively stable, some areas saw significant demographic shifts. These changes could be driven by gentrification, displacement, or migration patterns.



Percent Change in Educational Attainment (Bachelor's Degree or Higher and 25+)

- **Mean (5.13%):** There was a modest overall increase in educational attainment levels.
- **Median (4.52%):** The median aligns closely with the mean, suggesting a relatively balanced distribution of changes across tracts.
- **Standard Deviation (9.84%):** Some tracts saw substantial increases in educational attainment, while others experienced declines (min -26.13%).
- **25th percentile (-1.22%):** A quarter of the tracts saw slight decreases in educational attainment.
- **75th percentile (11.36%):** One-fourth of the tracts had strong gains in educational attainment.

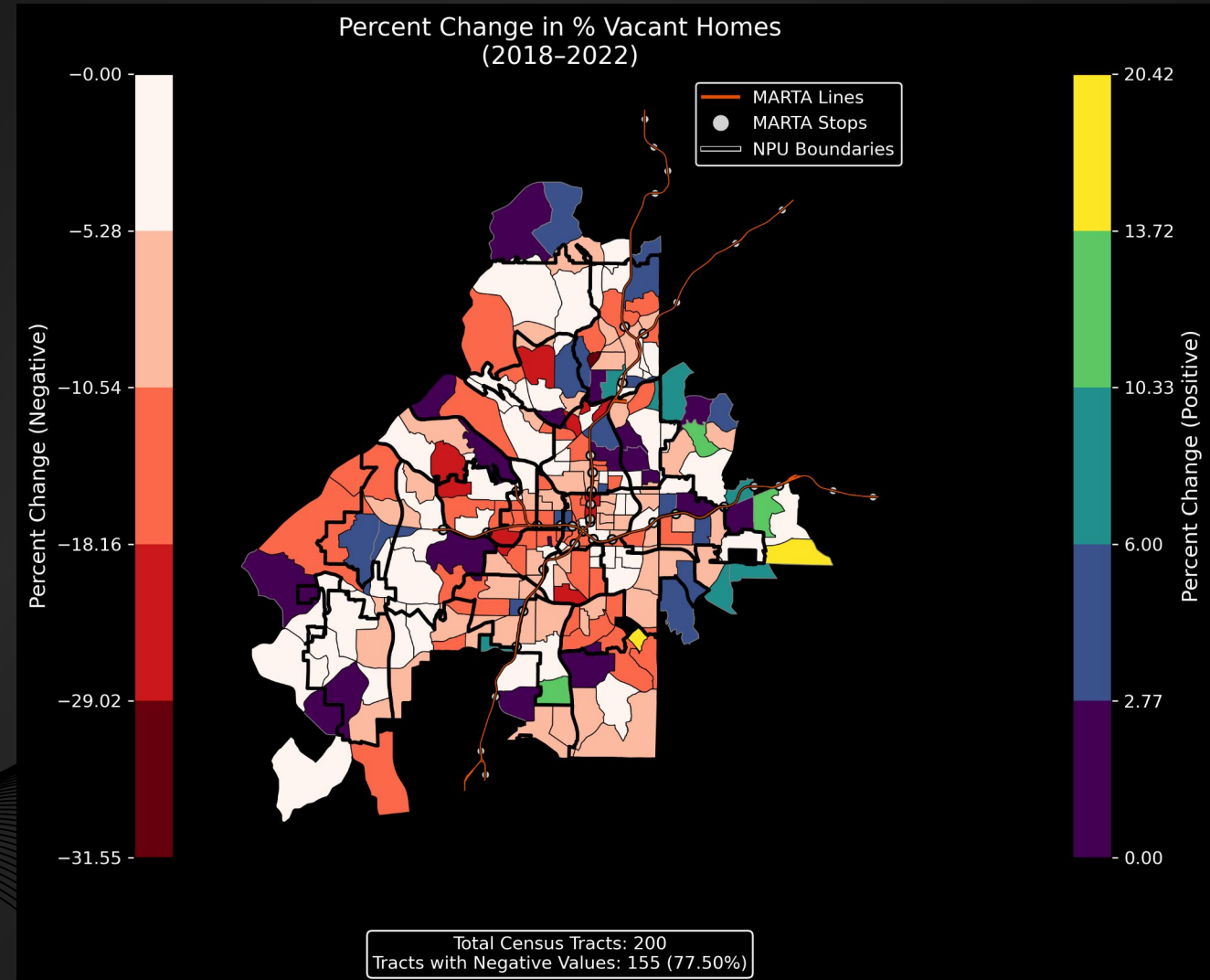
Key Takeaway: Educational attainment increased in many areas, likely reflecting new, more highly educated residents moving into these tracts. However, some areas saw declines, potentially tied to outmigration or socioeconomic changes.



Percent Change in % Vacant Homes

- **Mean (-5.33%):** On average, the percentage of vacant homes decreased, signaling potential improvement in housing utilization or market conditions.
- **Median (-5.56%):** The median is close to the mean, indicating a relatively even distribution of changes.
- **Standard Deviation (7.62%):** Some areas experienced significant decreases in vacancies, while others saw increases (max 20.42%).
- **25th percentile (-10.01%):** Many tracts saw notable reductions in vacancy rates.
- **75th percentile (-0.35%):** The majority of tracts had stable or decreasing vacancy rates.

Key Takeaway: The reduction in vacant homes suggests a stronger housing market overall, though some areas may still struggle with housing utilization or economic challenges.



To what extent is each % change explained by its 2018 “starting” value?

Including a basic summary statistic description of each distribution of deltas

D*mn*d from the start?

% Non-white

$r = -0.356$: A moderate negative correlation.

Census tracts with a higher initial percentage of nonwhite residents tend to experience smaller increases or decreases in the percentage of nonwhite residents. R-squared value **below 0.15**.

Educational Attainment

$r = -0.252$: A weak negative correlation.

Tracts with a higher starting percentage of bachelor's attainment tend to see slower growth in this metric. R-squared value **below 0.15**.

Median Home Value

$r = -0.458$: A moderately strong negative correlation.

Higher starting home values are linked to slower growth (or declines) in home values over time. R-squared (0.210), meaning ~21% of the variability is explained by the starting home value.

Median Household Income

$r = -0.295$: A weak-to-moderate negative correlation.

Tracts with higher initial household incomes tend to see smaller increases or decreases in income. R-squared value **below 0.15**.

% Vacant Homes

$r = -0.675$: A strong negative correlation.

Tracts with more vacant homes initially tend to experience sharper declines in the percentage of vacant homes. R-squared (0.456), meaning ~46% of the variability in percent change is explained by the starting percentage of vacant homes.

Starting-Line Predictors, Continued

% Vacant Homes shows the strongest relationship:

Tracts with higher starting vacancy rates tend to see sharp reductions, suggesting targeted improvements in housing vacancy.

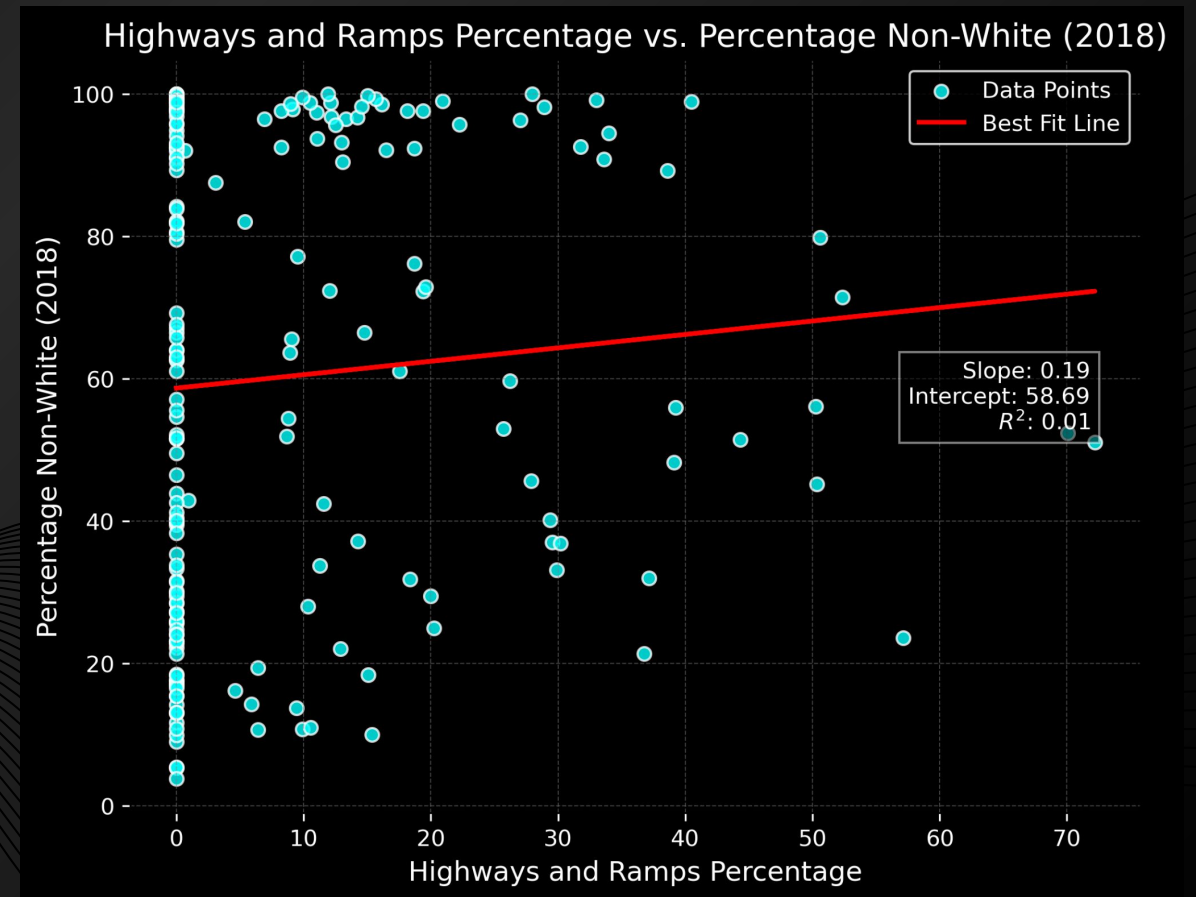
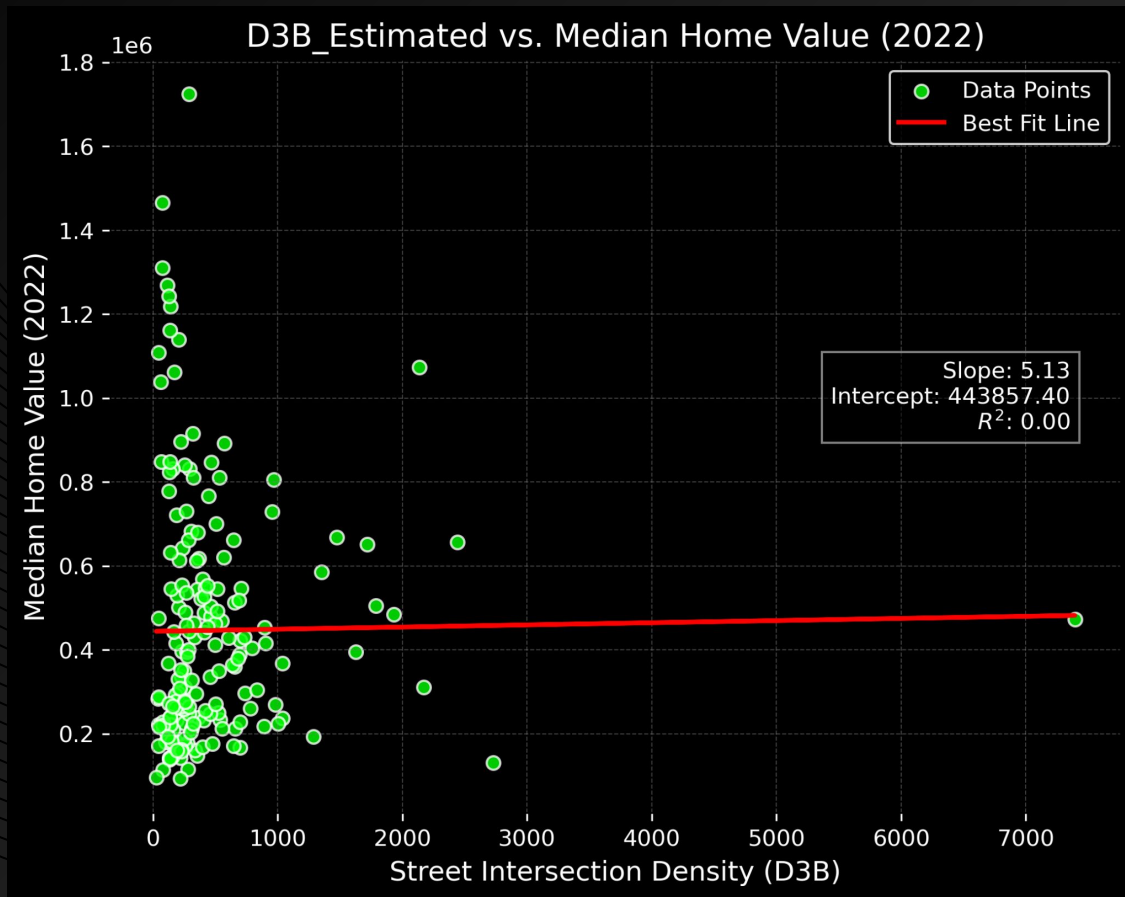
Median Home Value and Pct_Non_White also show moderate relationships:

This suggests that high starting home values or higher percentages of non-white residents are associated with slower changes over time.

Other Variables show weaker relations but statistically significant correlations:

This indicates that even small effects are consistent across the dataset.

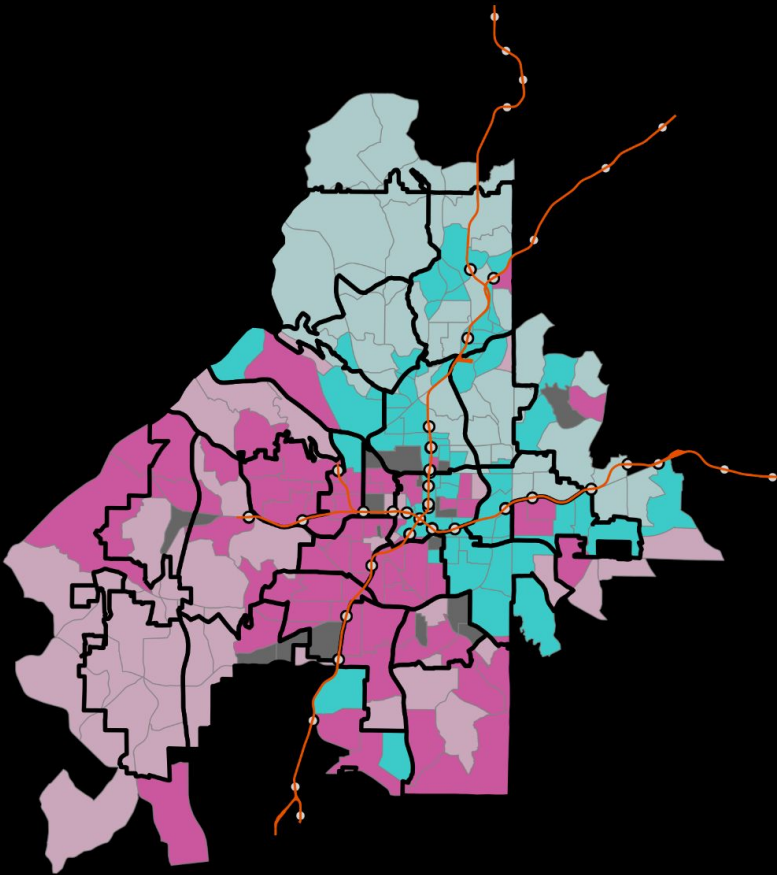
Additional Inconclusive Tropes, Assumption is their relationships were Canceled Out by Spatial Divergence



A detailed look at the cluster-profiles created by the K-Means classification

Summary stats are averages unless mentioned otherwise

Clustering of Atlanta Census Tracts by
Neighborhood Change

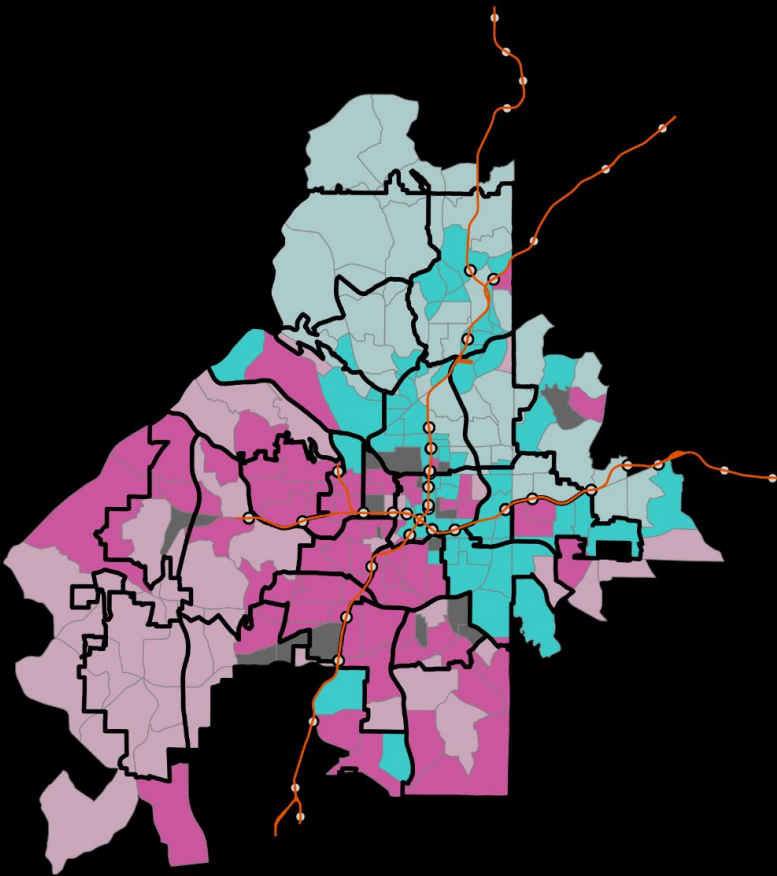


Wealthy, Stable Areas (Light Blue)

- **Median Home Value (2018):** \$648,088 (highest among clusters).
- **Home Value Percent Change:** 27.35% (modest increase—lowest among clusters).
- **Percent Non-White (2018):** 20.94% (lowest diversity).
- **Percent Non-White Change:** +1.16% (slight increase in diversity).
- **Vacancy Rate (2018):** 9.00% (low).
- **Vacancy Rate Change:** -0.88% (stable housing market).
- **Education Change:** +0.70% (minimal change in educational attainment).
- **Distance to MARTA Stop:** 1.07 miles (moderate transit access).

Interpretation: These are affluent, stable neighborhoods with limited demographic or economic change. The modest increases in home values and diversity suggest these areas are less susceptible to gentrification pressures.

Clustering of Atlanta Census Tracts by
Neighborhood Change

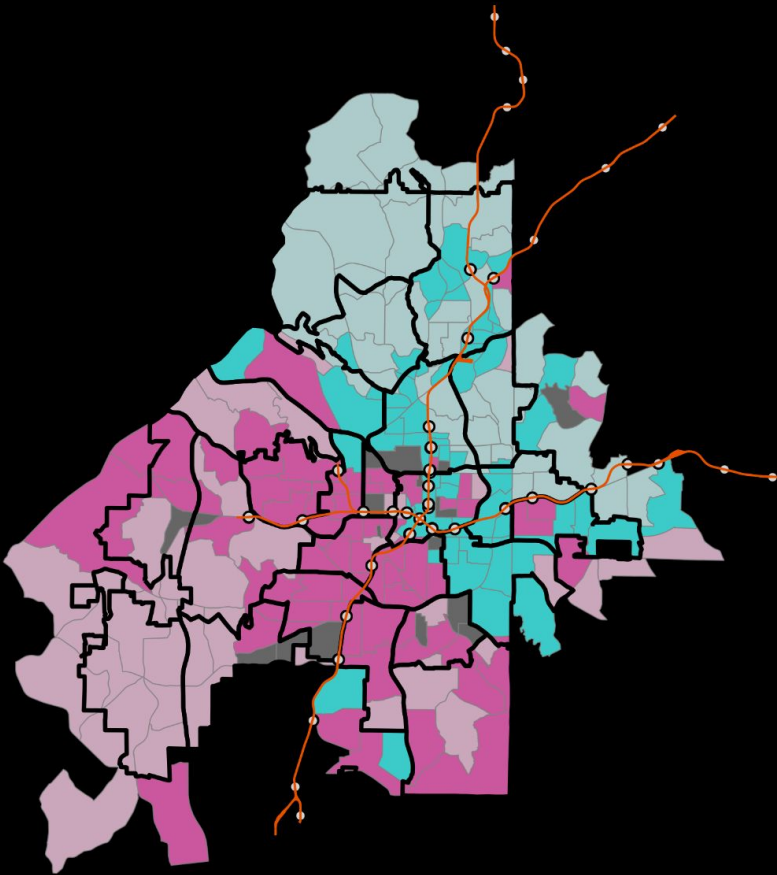


Middle-Class, Moderate Gentrification (Dark Blue)

- Median Home Value (2018): \$297,195 (mid-tier).
- Home Value Percent Change: 50.22% (moderate increase).
- Percent Non-White (2018): 40.04% (moderate diversity).
- Percent Non-White Change: +8.34% (increasing diversity).
- Vacancy Rate (2018): 17.01% (moderate).
- Vacancy Rate Change: -7.64% (substantial reduction—indicative of increased housing demand).
- Education Change: +3.07% (slight increase in residents with bachelor's degrees).
- Distance to MARTA Stop: 0.54 miles (good transit access).

Interpretation: These are middle-class areas experiencing gradual gentrification. The modest increase in educational attainment and home values, alongside reduced vacancies, points to steady, incremental change.

Clustering of Atlanta Census Tracts by
Neighborhood Change

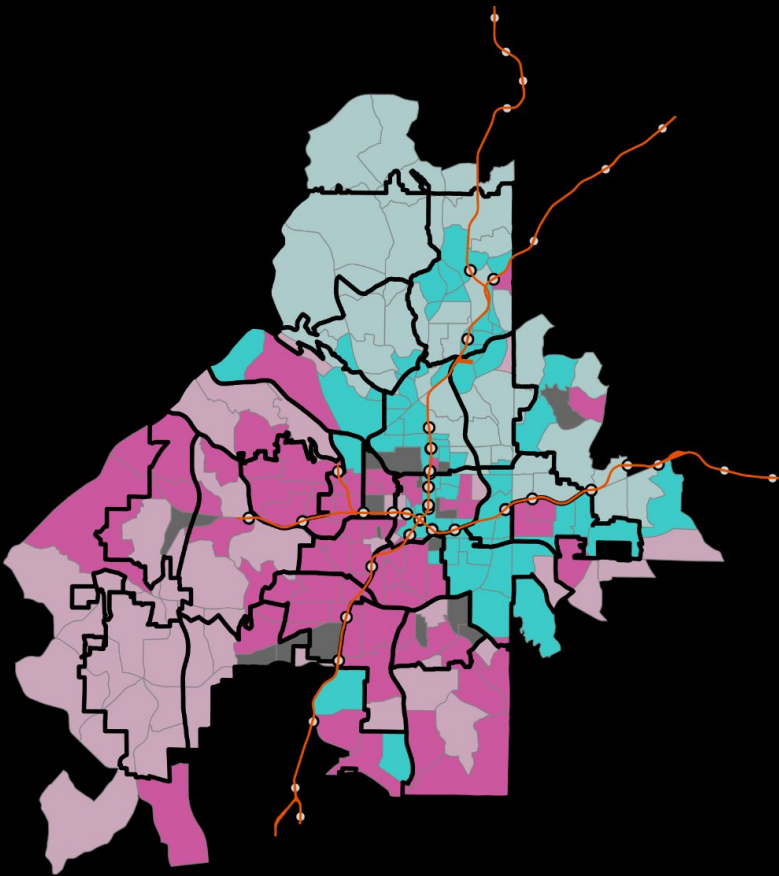


Historically Disadvantaged, Rapidly Gentrification (Dark Pink) → Base of MN Regression

- **Median Home Value (2018):** \$151,818 (low).
- **Home Value Percent Change:** 110.20% (highest increase—indicative of rapid gentrification).
- **Percent Non-White (2018):** 86.75% (predominantly non-White).
- **Percent Non-White Change:** -4.34% (decreasing diversity).
- **Vacancy Rate (2018):** 26.05% (high).
- **Vacancy Rate Change:** -10.01% (sharp reduction—indicative of high housing demand).
- **Education Change:** +9.54% (significant increase in bachelor's degree holders).
- **Distance to MARTA Stop:** 0.65 miles (good transit access).

Interpretation: These areas are undergoing rapid gentrification, with significant increases in home values and educational attainment and reductions in diversity and vacancies. The proximity to MARTA likely plays a role in this transformation.

Clustering of Atlanta Census Tracts by
Neighborhood Change

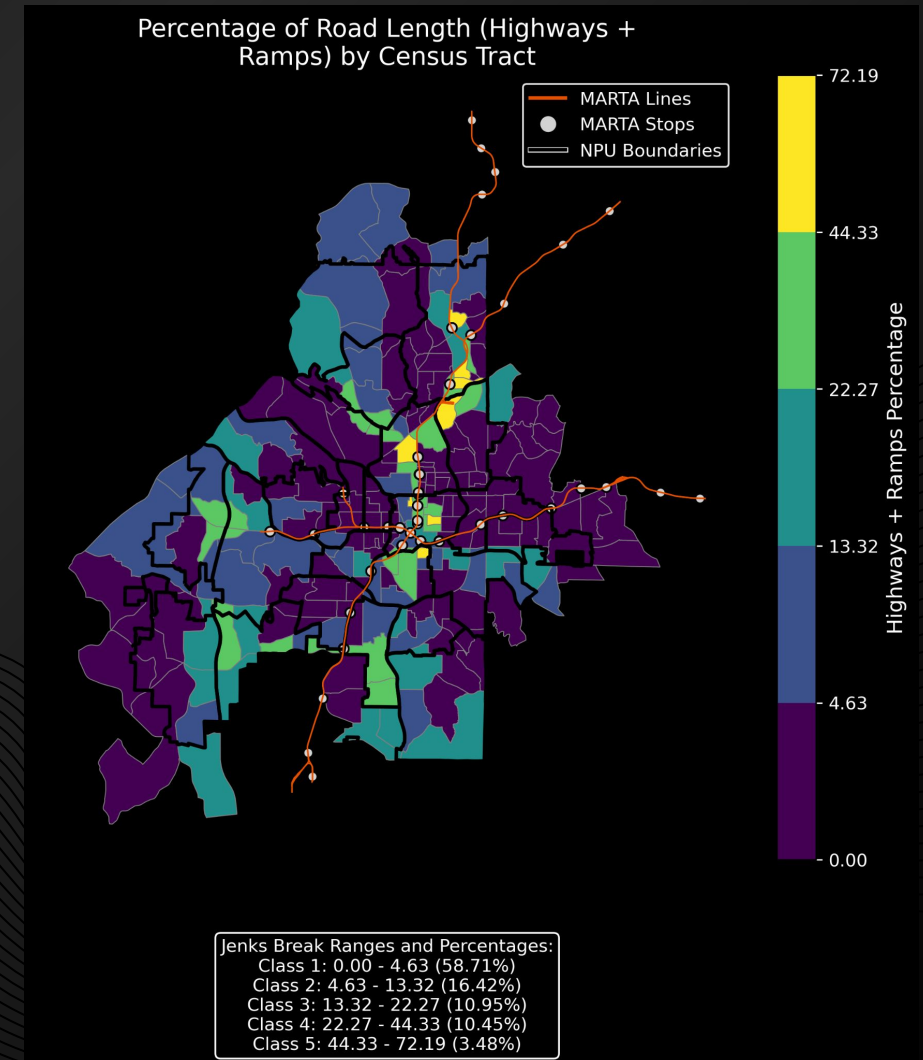
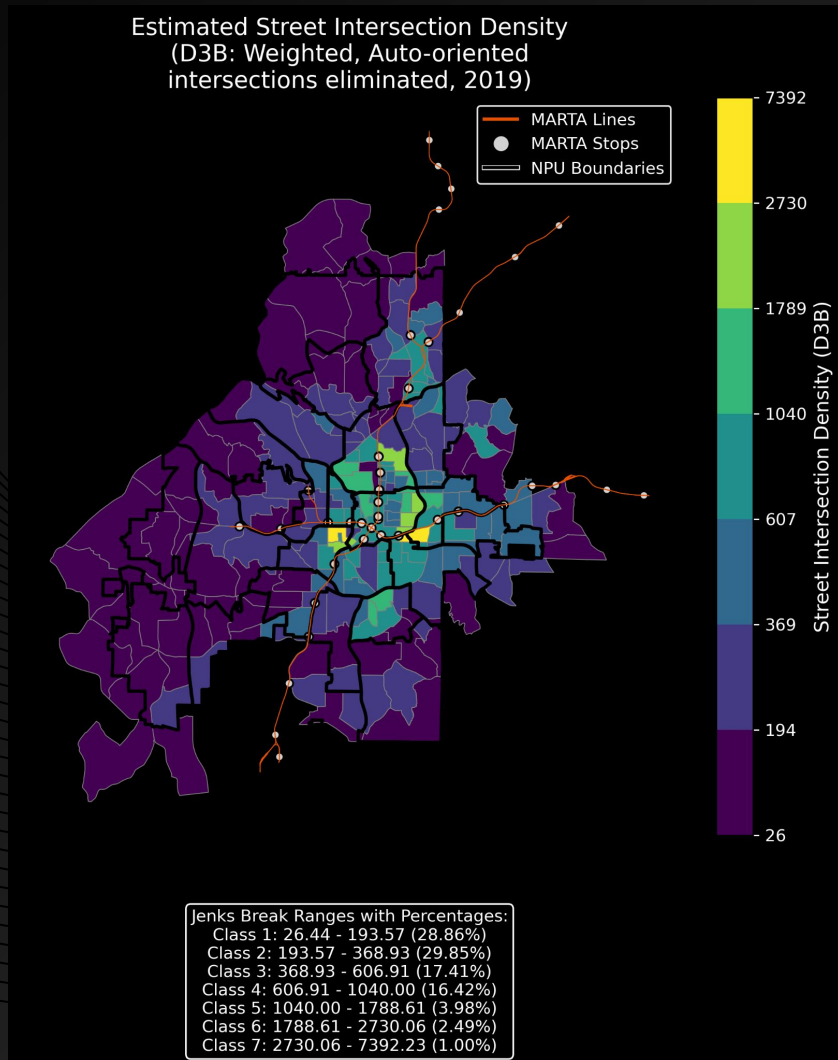


Low-Income, Slow Gentrification (Light Pink)

- Median Home Value (2018): \$157,278 (low).
- Home Value Percent Change: 80.89% (high increase).
- Percent Non-White (2018): 91.62% (predominantly non-White).
- Percent Non-White Change: -3.45% (slight decrease in diversity).
- Vacancy Rate (2018): 9.98% (low).
- Vacancy Rate Change: +0.71% (slight increase—indicative of weak housing demand).
- Education Change: +9.47% (significant increase in bachelor's degree holders).
- Distance to MARTA Stop: 2.03 miles (farthest from transit).

Interpretation: These areas show signs of limited gentrification. The increase in home values and educational attainment is offset by stagnant or slightly worsening vacancy rates and weaker transit access.

Reminder of Findings: “Bones” Variables



Multinomial Regression

Significant Predictive Variables

- **Proximity to MARTA:** This variable had a statistically significant relationship with some neighborhood change typologies, particularly when comparing wealthier, stable areas to historically disadvantaged, rapidly gentrifying areas. Tracts closer to MARTA were more likely associated with the latter, suggesting transit access may influence gentrification dynamics.
- **Intersection Density (D3B):** A negative relationship with wealthier and historically disadvantaged areas indicates that walkability, as measured by street intersection density, might differentiate between types of change but is not consistent across clusters.

Non-Significant Variables

- **Highways and Ramps Percentage:** Despite being a stable infrastructural characteristic, this variable showed limited or no statistical significance across most neighborhood types. This suggests that road infrastructure alone might not strongly correlate with the type of change, or its effects are context-dependent and require further exploration.

Multinomial Regression, Continued

Potential Issues

- **Insufficient Variables:** The model may need additional or better predictors. For example, socioeconomic data, historical trends, or land-use patterns might improve predictions.
- **Category Imbalance:** Some categories have very few observations (e.g., "Unclassified" with 7 instances), making it harder for the model to learn patterns.
- **Complexity of Neighborhood Change:** The phenomenon might not be well-captured by the three variables you included.
- **Overall accuracy:** 25%. Very low

Some classes (e.g., "Historically Disadvantaged, Rapid Gentrification") are predicted better than others, but no class performs particularly well.

Many classes (e.g., "Unclassified") have precision, recall, and F1-scores of 0, indicating the model struggles to predict them.

Non-Significant Variables

- **Distance_to_MARTA_Stop:**
 - Significant for Wealthy, Stable Areas (Cluster 1). Tracts further from MARTA are more likely to be classified here.
 - Weak or insignificant effects for other clusters.
- **D3B_Estimated:**
 - Significant negative relationship for Wealthy, Stable Areas (Cluster 1). Higher intersection density reduces the likelihood of this classification.
 - Marginally significant for Unclassified (Cluster 3), suggesting lower intersection density might push tracts here.
- **Highways_Ramps_Percentage:**
 - Not significant for any cluster but hints at slight negative relationships in several clusters.

Multinomial Regression, Continued

Model Limitations & Insights

- **Explained Variance:** While the model was statistically significant (p -value = 0.001), the pseudo R-squared value (~ 0.042) indicates that the included variables explain only a small portion of neighborhood change patterns. This suggests other unexamined factors, such as policy, market forces, or socio-cultural shifts, play a significant role.
- **Typology Classification Challenges:** Some neighborhood change categories, such as "Unclassified" or "Middle-Class, Moderate Gentrification," were harder to predict accurately, as reflected in the Scikit-learn precision and recall scores. This highlights potential overlap or ambiguity in the defined typologies.

Broader Implications

- While the analysis demonstrates that infrastructure variables like proximity to transit and walkability are relevant, **their predictive power alone is limited**. This underscores the importance of incorporating a broader set of socio-economic and temporal variables to achieve a more robust understanding of neighborhood change.
- Is it possible that **using a combination of 1 demographic variable and 1 infrastructure** variable has stronger predictive power? To what degree is infrastructure secondary to highly correlated demographic predictors of gentrification typologies? **This feels especially true when thinking about spatial divergence / convergence of similarities and differences on one variable alone.**
- Was it right to use the most aggressively gentrifying typology as the **"reference category?"**

Next Steps / Further Exploration

Panel Analysis

Analytic capabilities and options for predictive power might be improved by including more 5-year time slices for a panel analysis.

Apportionment to NSA / NPU

Cautiously, I could employ a polygon apportionment that creates values at the NSA and / or NPU geometries. Finding a way to including weighting in this apportionment would improve its reliability.

Dasymetric Weighting

Weighting based on distribution of land use, population, housing units, or the full street network can be used to make the walkability index – and the apportionment suggested for future use to the left – more reliable.

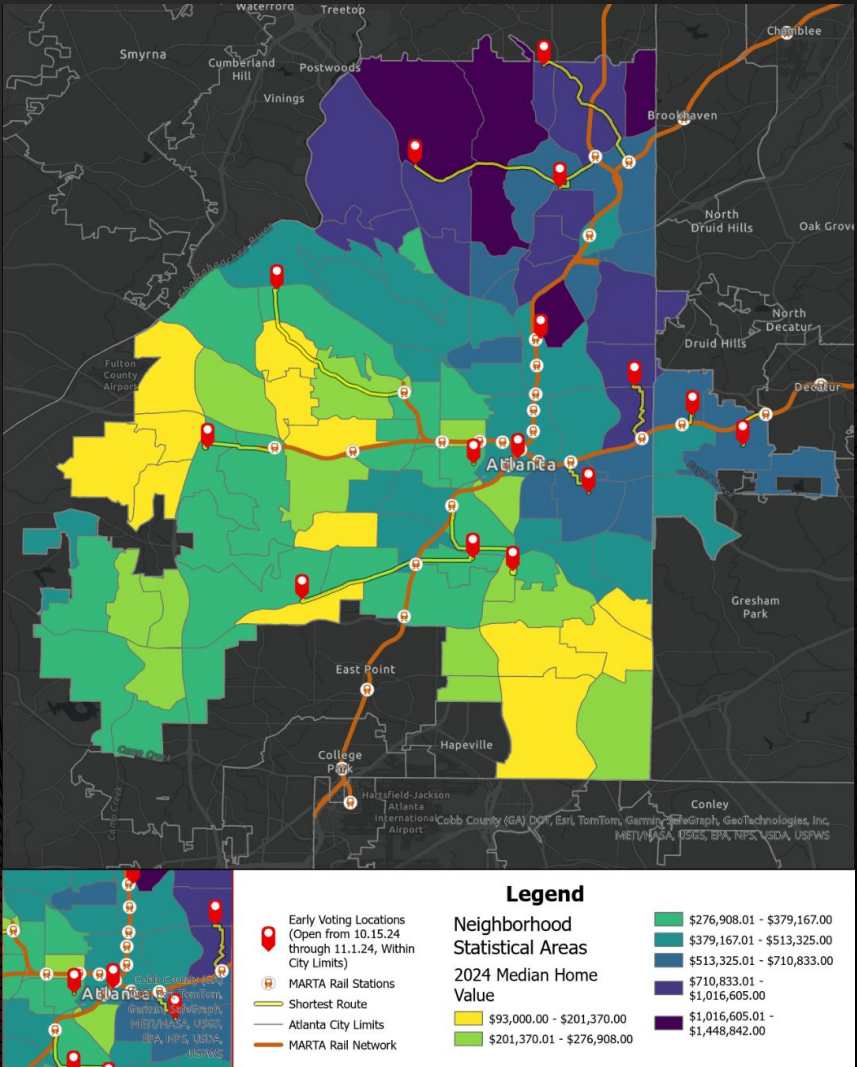
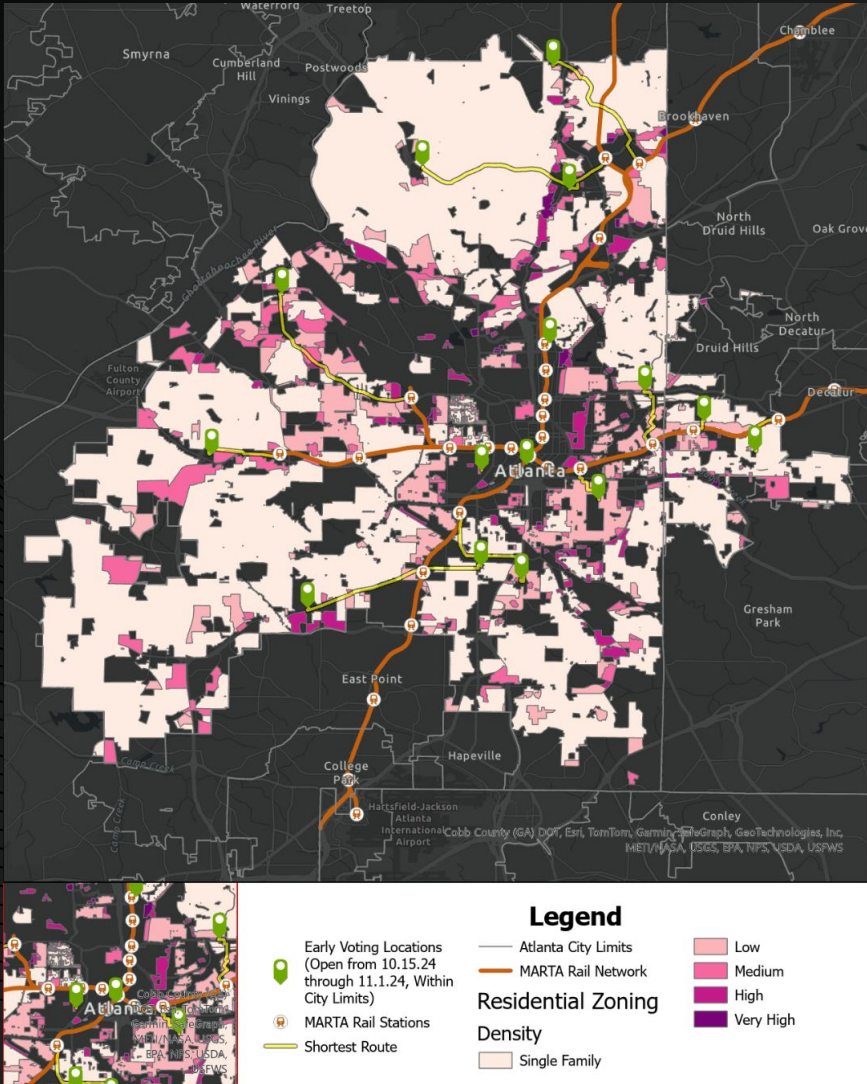
Additional Variables

The findings lead me to cautiously believe that looking at more variables over time and continuing to mix, match, and add them in the multinomial regression will help. Does this feel true? I'm also aware adding too many variables will muddy the predictive power or create noise.

Include POI Data

Accessibility to the Beltline is likely a key driver, as we know from our research, so pulling in Beltline Access points may aid as a powerful local-knowledge based predictor.

Spatial Divergence / Convergence, Continued



Data Sources

- Crosswalk File and Weighting: Steven Manson, Jonathan Schroeder, David Van Riper, Katherine Knowles, Tracy Kugler, Finn Roberts, and Steven Ruggles. IPUMS National Historical Geographic Information System: Version 19.0 [dataset]. Minneapolis, MN: IPUMS. 2024. <http://doi.org/10.18128/D050.V19.0>
- Atlanta Regional Commission (Neighborhood Statistical Areas and NPU's)
- Open Street Map (drive network by census tracts, MARTA stops, and MARTA Rail)
- EPA National Walkability Index Data Set (Smart Location Indicators, <https://catalog.data.gov/dataset/walkability-index7>)
- Urban Displacement Project GitHub and Open Methodology, <https://github.com/urban-displacement/displacement-typologies/tree/main/code>



Thank you!

Do you have any questions?